

Genotipado con microarrays

Máster en Sistemas Inteligentes

Luis Quintales
Departamento de Informática



VNIVERSIDAD
DSALAMANCA

- Patrones de herencia
- Microarray de genotipado
- Bases de datos de variantes
- Direct-to-Consumer Genomic Testing
- Personal Genome Project
- Enfermedades monogénicas
 - Alzheimer y APOE
 - Cáncer de mama y útero y BRCA1, BRCA2
 - Fibrosis quística y CFTR
- Interpretación: Odds ratio



OMIM®
Online Mendelian Inheritance in Man®
An Online Catalog of Human Genes and Genetic Disorders
Updated March 9, 2021

Search OMIM for clinical features, phenotypes, genes, and more...

Advanced Search : [OMIM](#), [Clinical Synopses](#), [Gene Map](#)
Need help? : [Example Searches](#), [OMIM Search Help](#), [OMIM Video Tutorials](#)
Mirror site : <https://mirror.omim.org>

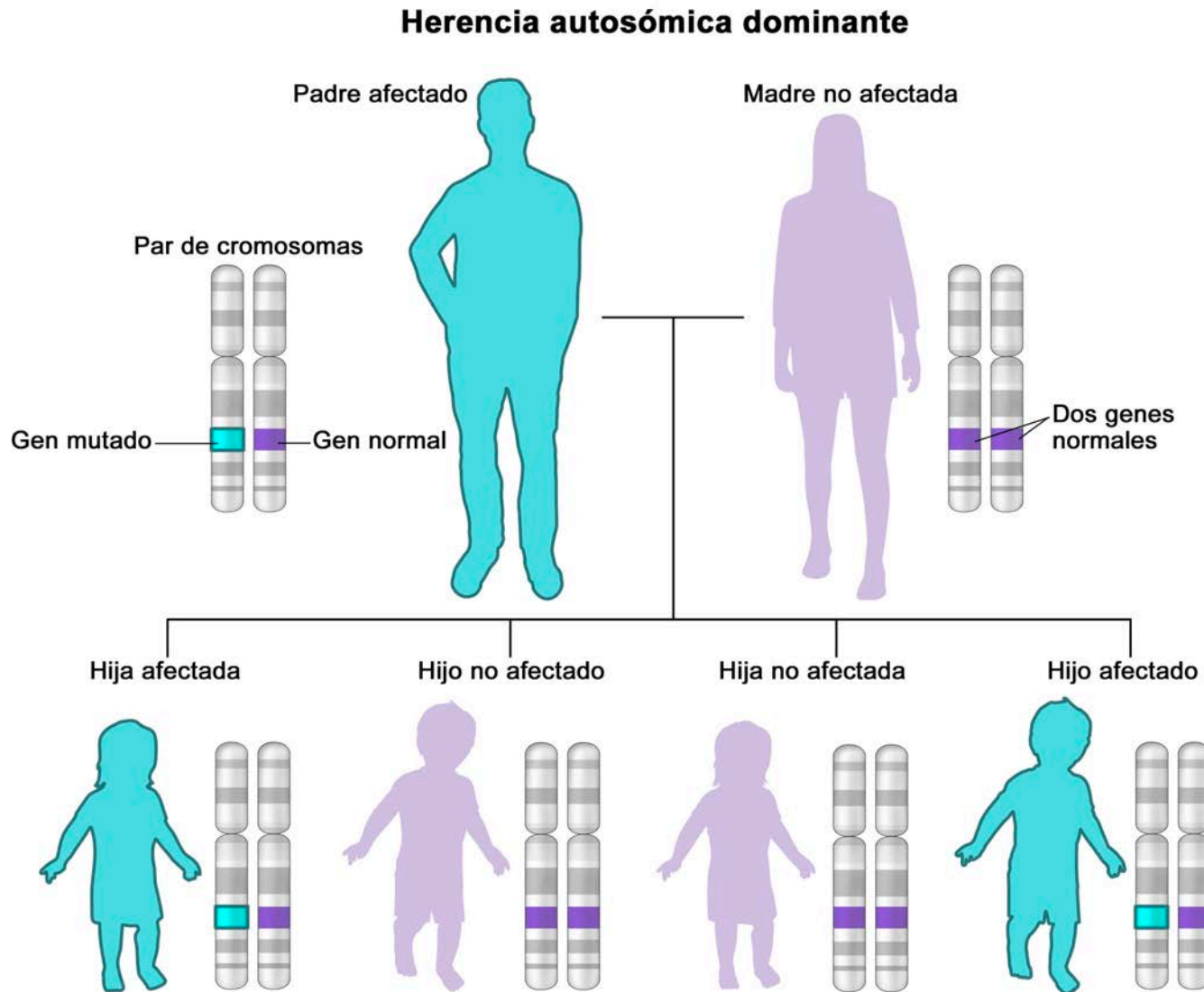
OMIM is supported by a grant from NHGRI, licensing fees, and generous contributions from people like you.

OMIM Entry Statistics

Number of Entries in OMIM (Updated March 9th, 2021) :

MIM Number Prefix	Autosomal	X Linked	Y Linked	Mitochondrial	Totals
Gene description *	15,597	747	51	37	16,432
Gene and phenotype, combined +	28	0	0	0	28
Phenotype description, molecular basis known #	5,626	355	5	34	6,020
Phenotype description or locus, molecular basis unknown %	1,412	112	4	0	1,528
Other, mainly phenotypes with suspected mendelian basis	1,657	102	3	0	1,762
Totals	24,320	1,316	63	71	25,770

Herencia autosómica dominante

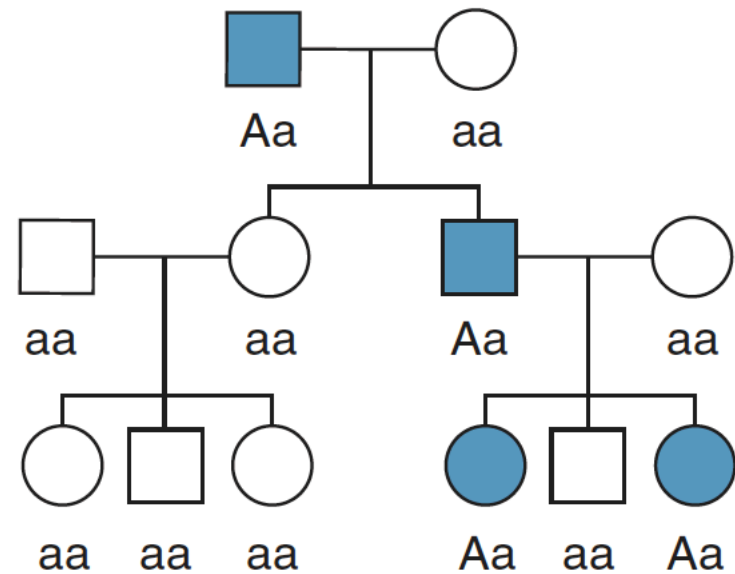


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Herencia autosómica dominante

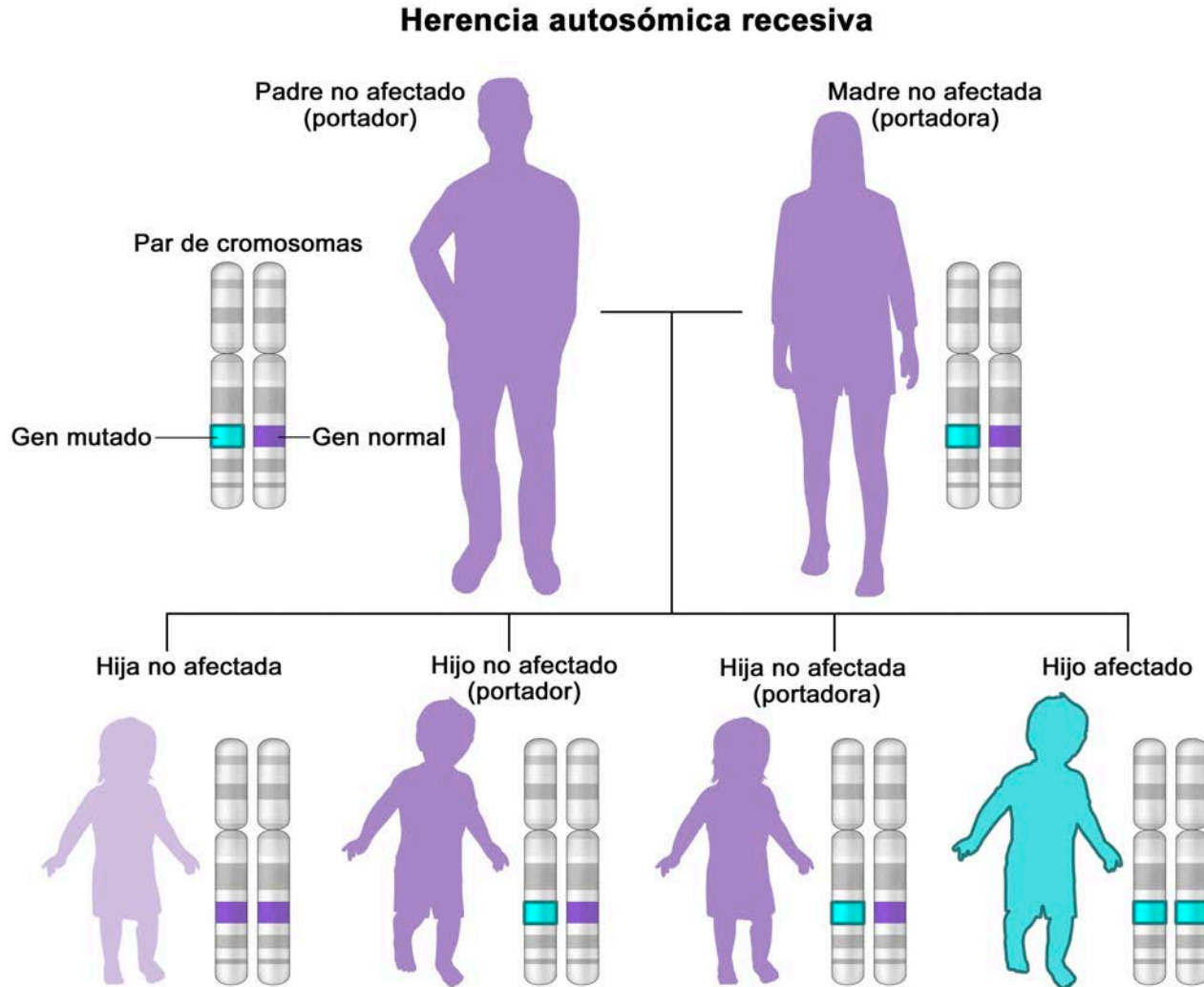
		Unaffected parent	
		a	a
Affected parent	A	Aa	Aa
	a	aa	aa

Cuadro de Punnet



Ejemplo de genealogía

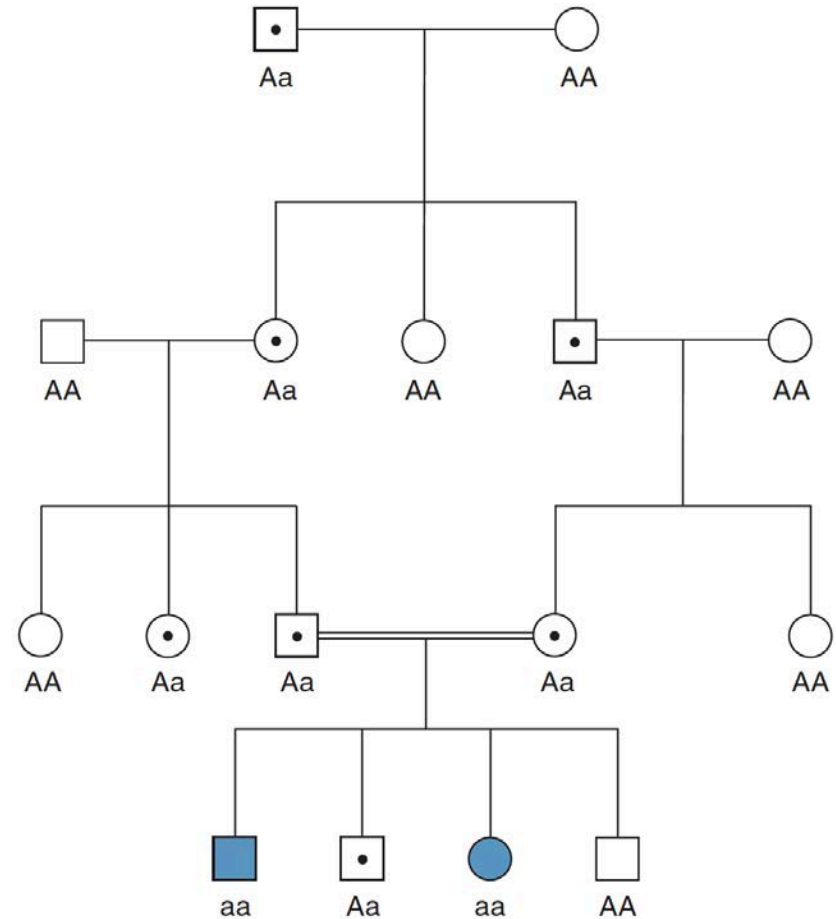
Herencia autosómica recesiva



Herencia autosómica recesiva

		Carrier parent	
		A	a
Carrier parent	A	AA	Aa
	a	Aa	aa

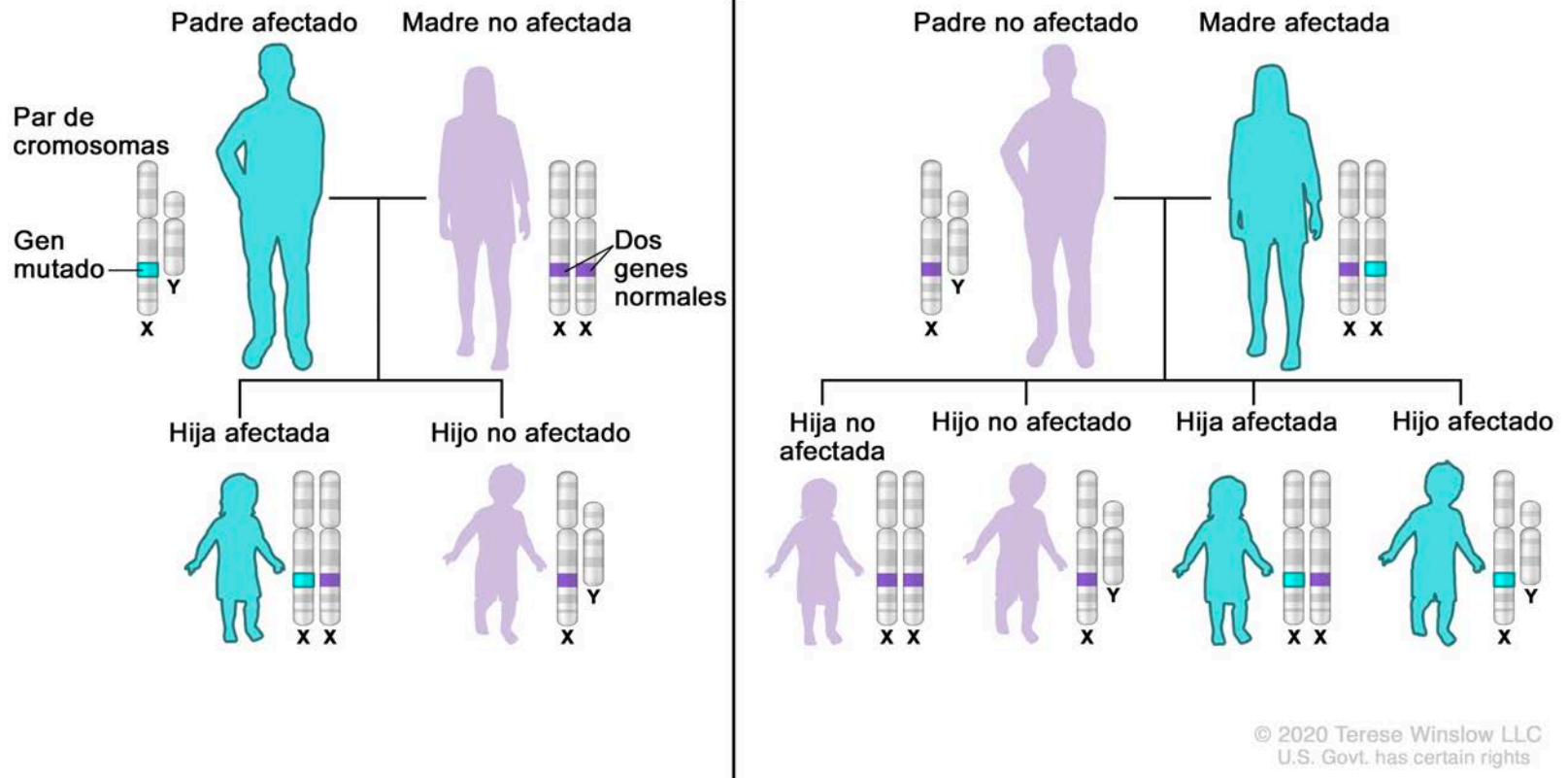
Cuadro de Punnet



Ejemplo de genealogía

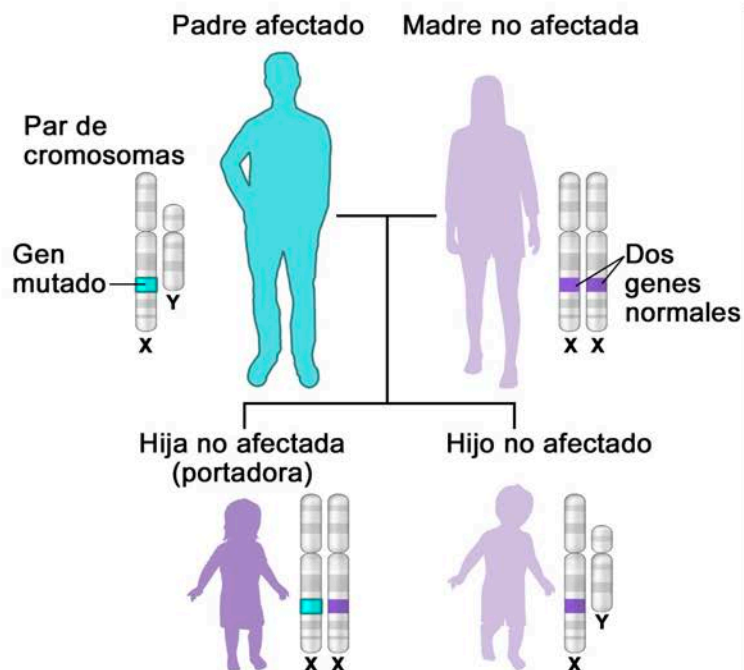
Herencia dominante ligada al cromosoma X

Herencia dominante ligada al cromosoma X



Herencia recesiva ligada al cromosoma X

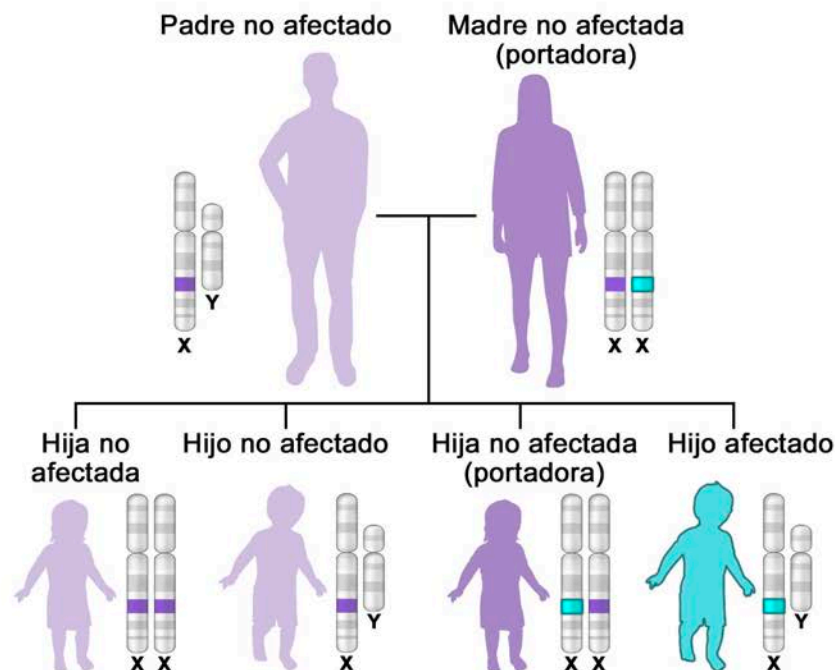
Herencia recesiva ligada al cromosoma X



		Mother	
		X_1	X_1
Father	X_2	X_2X_1	X_2X_1
	Y	X_1Y	X_1Y

Daughters: 100% carriers

Sons: 100% normal



		Mother	
		X_1	X_2
Father	X_1	X_1X_1	X_1X_2
	Y	X_1Y	X_2Y

Daughters: 50% normal, 50% carriers

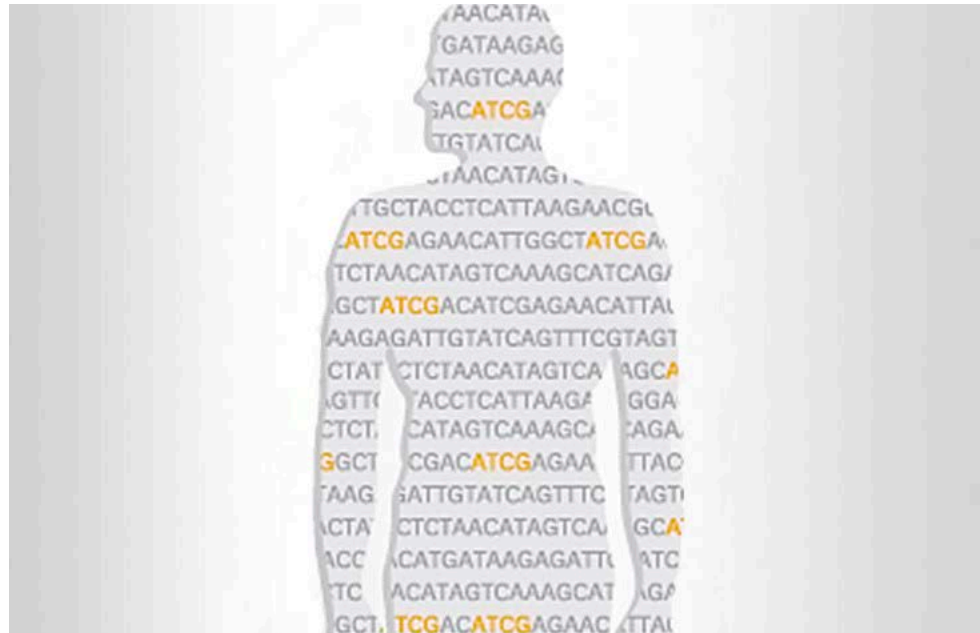
Sons: 50% normal, 50% affected

slow LLC
in rights

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Whole-Genome Genotyping

- Microarrays
- NGS

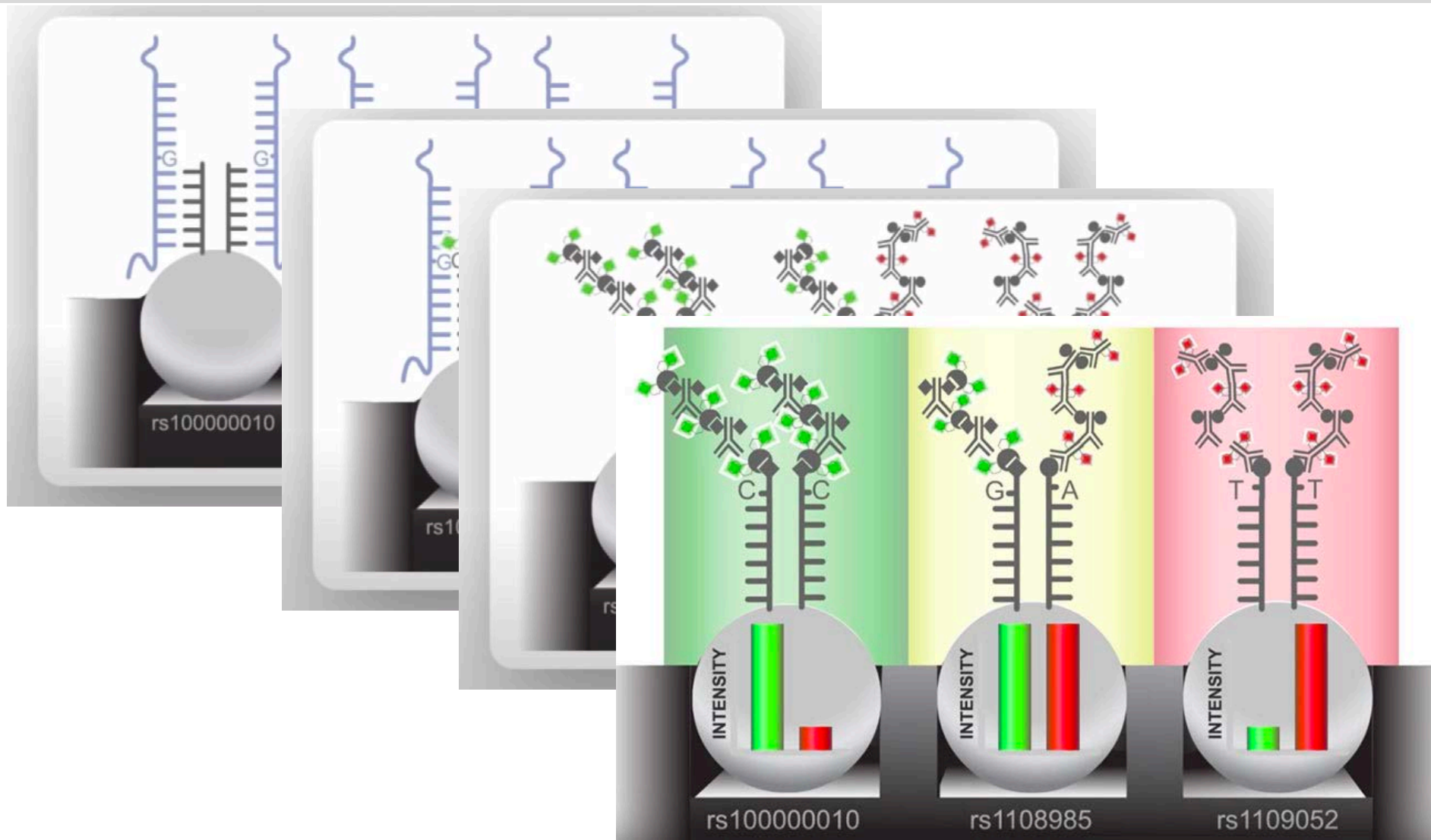


Illumina Infinium OmniExpress-24 Kit

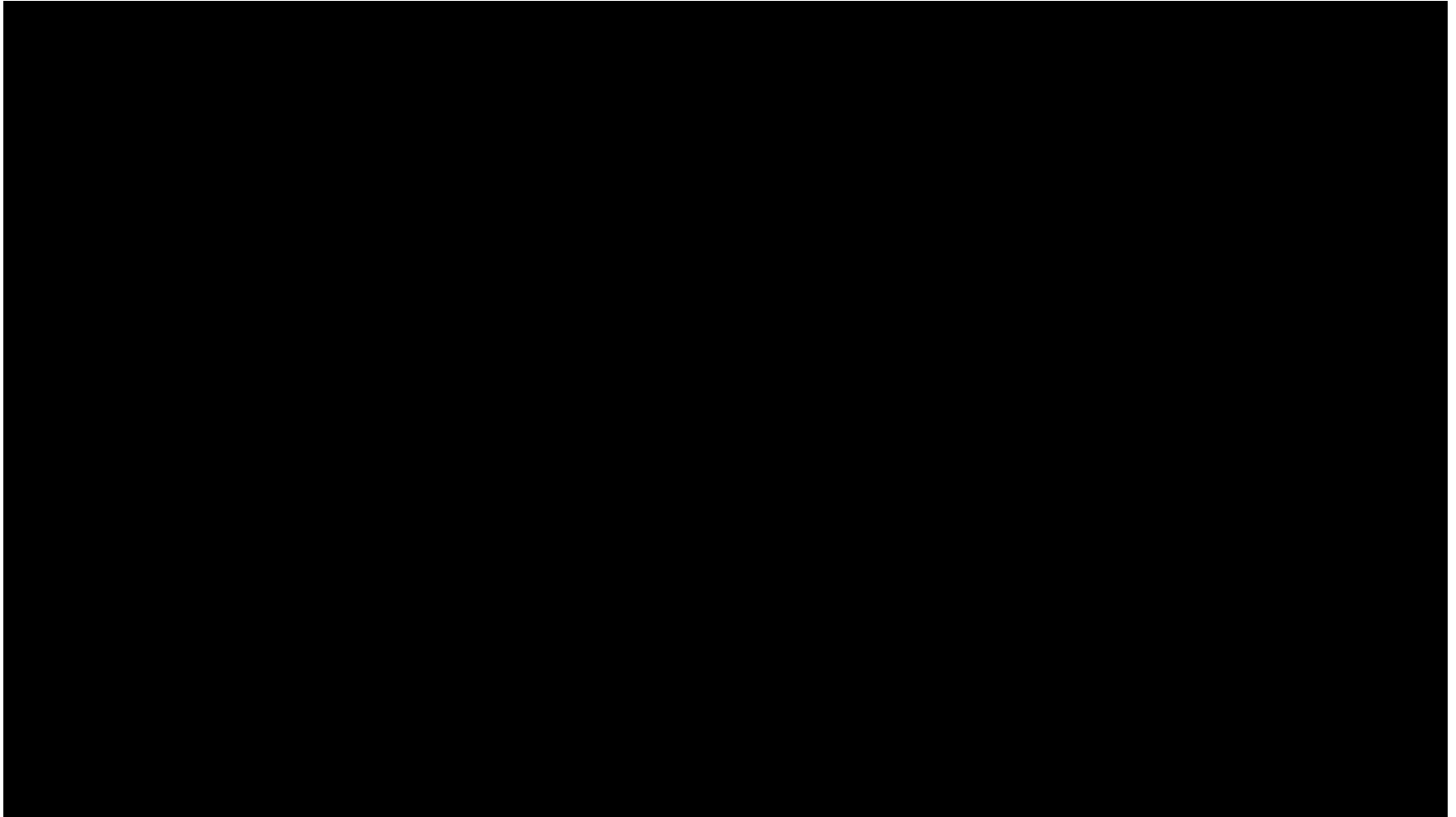


24 muestras por array
Sondas: aprox. 710.000

Illumina Infinium Assay



Illumina Infinium Assay



Práctica: Illumina microarrays

- ¿Qué información hay en un microarray de Illumina?
- ¿Cuántas sondas/SNPs incorpora?
- ¿Cuántas sondas hay para alguno de los genes que pueden interesarnos?

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dbSNP Reference (rs) number

- El catálogo RefSNP es una colección no redundante de variantes publicadas que han sido agrupadas, integradas y anotadas
- El número RefSNP es un método de acceso estable independientemente de las diferencias en los ensamblajes genómicos
- Los números RefSNP facilitan los estudios a gran escala en genética de asociación, genética médica, genómica funcional y farmacológica, genética de poblaciones y biología evolutiva, genómica personal y medicina de precisión
- Proporcionan una notación de variantes estable para el análisis de mutaciones y polimorfismos, la anotación, la elaboración de informes, la extracción de datos y la integración de datos

Práctica: dbSNP y bases de datos asociadas

- dbSNP
- ClinVar
- LitVar
- Variation Viewer
- OMIM

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Direct-to-Consumer Genomic Testing (23andMe)

23andMe[®] COVID-19 STUDY SHOP ▾ LEARN ▾ SIGN IN REGISTER KIT HELP Shop

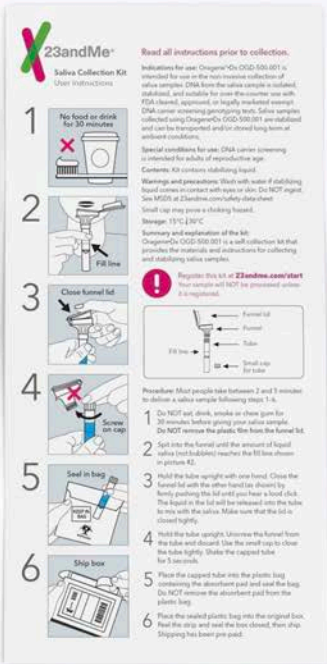
What's in the kit.



saliva collection kit



specimen bag



step by step instructions



funnel lid

saliva collection tube

tube cap

tube container

Direct-to-Consumer Genomic Testing (MyHeritage)

DNAFamily treeResearchHelp

Our technology and standards

In accordance with the industry standard, MyHeritage uses the Illumina OmniExpress-24 chip, with hundreds of thousands of strategically selected probes to capture the greatest amount of genetic variation, and facilitate research. The lab is CLIA-certified (CLIA is the Clinical Laboratory Improvement Amendments), a standard indicating, among other key qualifications, that all lab technicians who analyze MyHeritage users' DNA have achieved a particular level of qualification. The lab is also CAP-accredited, the highest level of credential in the United States for this kind of laboratory.



Descarga de datos 23andMe

HOMEANCESTRY & TRAITSHEALTHRESEARCHFAMILY & FRIENDSBuy kits

BrowseDownload

Download Raw Data

As part of your 23andMe service, we provide full access to your raw genetic data to browse or download and do with as you please. Before downloading, however, you should be aware that there are some risks that you should know about.

Important warnings to consider:

If you decide to download your raw genetic data from 23andMe, **take caution in uploading your data to any third party applications or any service outside of your 23andMe account.** These include services that offer to provide a health interpretation of your raw data, in addition to public genealogy websites.



Your raw data is not validated for accuracy. It has undergone a general quality review, however only select data (which are included in the Health reports we provide) have been individually validated for accuracy. Raw data should not be used for medical purposes and we do not recommend the use of third party services that claim to interpret raw data to provide health information.



Your raw data could include sensitive information. By downloading your raw data, you could discover sensitive health information about yourself or family members with whom you share DNA.



Once you download your raw data, we cannot ensure it will be kept secure. We secure your data on 23andMe servers using advanced encryption and other technical, physical, and procedural safeguards. Once you download your data, we cannot guarantee the security of that downloaded data file.

Descarga de datos en MyHeritage

Manejo de los kits de ADN

Kit de ADN	Tipo	Estado	
 Asignado a usted MH-TQ3L62	Kit de ADN de Salud de MyHeritage	Los resultados están listos	⋮
 Comprado por usted MH-V3PN73	Kit de ADN de Salud de MyHeritage	Los resultados es	
 Asignado a L. BG MH-46WX72	Kit de ADN de Salud de MyHeritage	Los resultados es	

Descargar kit

Ver Estimación Étnica

Coincidencias de ADN

Ver reporte de salud

Borrar kit

Descarga de archivo de datos de ADN

El archivo de datos de ADN incluye los datos sin procesar de su muestra de ADN producidos por el laboratorio. La prueba de ADN de MyHeritage cubre entre 600.000 y 700.000 SNPs (polimorfismos de un solo nucleótido) usados para calcular sus resultados y compararlos con los resultados de ADN de otras personas.

Usted puede leer más acerca de los SNPs en el [International Society of Genetic Genealogy Wiki](#).

Su archivo de datos de ADN se descargará como un archivo de texto. Los datos aparecen en una tabla, en la que cada línea representa un SNP. Incluye los siguientes valores:

- rsID - Reference SNP Cluster ID. Este número es la escala de fiabilidad que permite el control del conocimiento sobre los SNPs contribuidos por diferentes proyectos de investigación en todo el mundo sin duplicaciones ni solapados. Cada SNP está asociado con un rsID.
- Cromosoma - Especifica en qué cromosoma está localizado el SNP específico.
- Posición - Indica la posición del SNP en la parte superior del cromosoma. Cada cromosoma está compuesto por dos áreas en la estructura en hélice llamada un alelo.
- Alelo 1 - Cuál nucleótido de su secuencia de ADN presenta en esta posición en el cromosoma que es una estructura sencilla de la hélice.
- Alelo 2 - Cuál nucleótido de su secuencia de ADN presenta en esta posición en el cromosoma que es una estructura doble de la hélice.

Usted puede leer más sobre nucleótidos y la estructura de doble hélice del ADN en el [International Society of Genetic Genealogy Wiki](#).

Un ejemplo del archivo de datos de ADN tiene esta apariencia:

```
RSID, CHROMOSOME, POSITION, RESULT
"rs3094315", "1", "752566", "AA"
"rs3131972", "1", "752721", "AA"
"rs12562034", "1", "768448", "GG"
"rs12124819", "1", "776546", "AA"
"rs11240777", "1", "798959", "GG"
"rs6681049", "1", "800007", "CC"
```

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Personal Genome Project

Personal Genome Project: Global Network

Projects ▾ News Contact

The Personal Genome Project

The Personal Genome Project, initiated in 2005, is a world dedicated to creating public genome, health, and scientific progress, but has been hampered by the lack of willing participants to publicly share their data.



**The Harvard
Personal Genome Project**
we are open for science

About Participate Use Data Donate Participant login

International Projects

The Global Network of Personal Genome Projects includes:

• Harvard PGP (United States)

Founded in August 2005, the Harvard Personal Genome Project is based at the Harvard Medical School.

[Go to the Harvard PGP website](#)

• PGP Canada (Canada)

Founded in December 2012, PGP Canada is operated by the Hospital for Sick Children.

[Go to PGP Canada website](#)

• PGP UK (United Kingdom)

Founded in November 2013, PGP UK is led by Stephen

[Go to the PGP UK website](#)

• Genom Austria (Austria)

Founded in November 2014, Genom Austria is based at the

[Go to Genom Austria website \(German language\)](#)

• PGP China (People's Republic of China)

Announced in October 2017, The Personal Genome Project is now collecting contact information from interested participants.

[Go to the PGP China website \(in Chinese and English\)](#)

PGP GENOME AND TRAIT DATASETS

GENOME DATA

In addition to whole genome sequencing, the Harvard PGP has a variety of other data types (including but not limited to performed genomes and exomes to direct-to-consumer genotyping).

GENOME REPORTS

To provide some insight into whole genome sequences, the Harvard PGP has generated genome reports. Genome reports generated by GET-Evidence

TRAIT AND SURVEY DATA

The Harvard PGP data is uniquely valuable because it combines trait and health data with genomic data. Our trait surveys query participants regarding 239 self-reported

PARTICIPANT PROFILES

Participant profiles present all public data associated with a participant, providing a more complete of a participant: it is a form of biological profiling that captures important aspects in human health.

MICROBIOME DATA

Microbiome profiles investigate the types of bacteria in and on a participant. A microbiome profile is a more complete of a participant: it is a form of biological profiling that captures important aspects in human health.

Personal Genome Project

Log in ►

Public data ▾ About ▾

Public genetic data

See also:

- [Genetic data statistics](#) — published data over time, split by data provider
- [Participant directory](#) — all enrolled users

All data types Search

Participant	Published	Data type	Source	Name	Download	Report
huC07Z70	2021-02-15	genetic data - AncestryDNA	Participant	AncestryDNA.JL	Download (17.1 MB)	
hu278AF5	2021-01-24	23andMe	Participant	23andMe	Download (15.9 MB)	
huA4D9DC	2020-09-24	Ancestry DNA	Participant	Ancestry DNA	Download (5.62 MB)	
huEB5501	2020-09-19	Ancestry DNA genetic data	Participant	Jam's Ancestry DNA Raw Data	Download (17.5 MB)	
hu2F3EC9	2020-08-20	Family Tree DNA	Participant	Family Tree DNA Family Finder Build 37 Autosomal Raw Data	Download (5.34 MB)	
huF7A4DE	2020-08-06	nebula genomics bam file	Participant	J's nebula genomics bam file	Download (38.9 GB)	
huF7A4DE	2020-08-03	nebula genomics data vcf	Participant	vcf from nebula genomics	Download (341 MB)	
huE29BED	2020-08-01	23andMe	Participant	23andMe genotyping	Download (4.96 MB)	
hu60B840	2020-07-12	23andMe	Participant	Genotyping data	Download (16.1 MB)	
huAB27F3	2020-07-06	Family Tree DNA	Participant	FTDNAAM	Download (14.9 MB)	
hu415DAA	2020-07-01	Veritas Genetics	Participant	hu415DAA_MyGenome	Download (668 MB)	
hu105A29	2020-06-21	VCF from Dante Labs vs GRCh37 (gz)	Participant	VCF from Dante Labs vs GRCh37	Download (246 MB)	
huC50E34	2020-06-12	Ancestry VCF	Participant	European100	Download (5.48 MB)	
hu5CA617	2020-06-12	Veritas Genetics	Participant	HopperZIP	Download (673 MB)	
hu80D170	2020-06-09	23andMe	Participant	Genome Data 23 and me	Download (14.7 MB)	
hu28CF73	2020-05-27	23andMe	Participant	Jeff_Hau_23_and_me	Download (5.60 MB)	

Descargar el genoma humano (UCSC)



UNIVERSITY OF CALIFORNIA
SANTA CRUZ Genomics
Institute



UCSC

Genome Browser

[Home](#) [Genomes](#) [Genome Browser](#) [Tools](#) [Mirrors](#) [Downloads](#)



Our tools

- **Genome Browser**
interactively view genomic data
- **COVID-19 RefSeq**
use the SARS-CoV-2 RefSeq
- **BLAT**
rapidly align sequences
- **Table Browser**
download data from the browser
- **Variant Annotator**
get functional effects
- **Data Integrator**
combine data sources
- **Genome Browser**
run the Genome Browser
- **In-Silico PCR**
rapidly align PCR primers
- **LiftOver**
convert genome coordinates
- **Track Hubs**
import and view data
- **REST API**
returns data in JSON

[More tools...](#)

Human genome

Dec. 2013 (GRCh38/hg38)

- [Genome sequence files and select annotations \(2bit, GTF, GC-content, etc\)](#)
- [Sequence data by chromosome](#)
- [Annotations](#)
- [SNP-masked fasta files](#)
- [LiftOver files](#)
- [Pairwise alignments](#)
- [Multiple alignments](#)
- [Patches](#)

Feb. 2009 (GRCh37/hg19)

- [Genome sequence files and select annotations \(2bit, GTF, GC-content, etc\)](#)
- [Sequence data by chromosome](#)
- [Annotations](#)
- [GC percent data](#)
- [Protein database for hg19](#)
- [SNP-masked fasta files](#)
- [LiftOver files](#)
- [Pairwise alignments \(primates\)](#)
- [Pairwise alignments \(other mammals\)](#)
- [Pairwise alignments \(other vertebrates\)](#)
- [Multiple alignments](#)
- [Patches](#)

Mar. 2006 (NCBI36/hg18)

- [Data and annotations](#)

May 2004 (NCBI35/hg17)

- [Data and annotations](#)

Jul. 2003 (NCBI34/hg16)

- [Data and annotations](#)

Apr. 10, 2003 (hg15)

- [Data and annotations](#)

Nov. 14, 2002 (hg13)

- [Data and annotations](#)

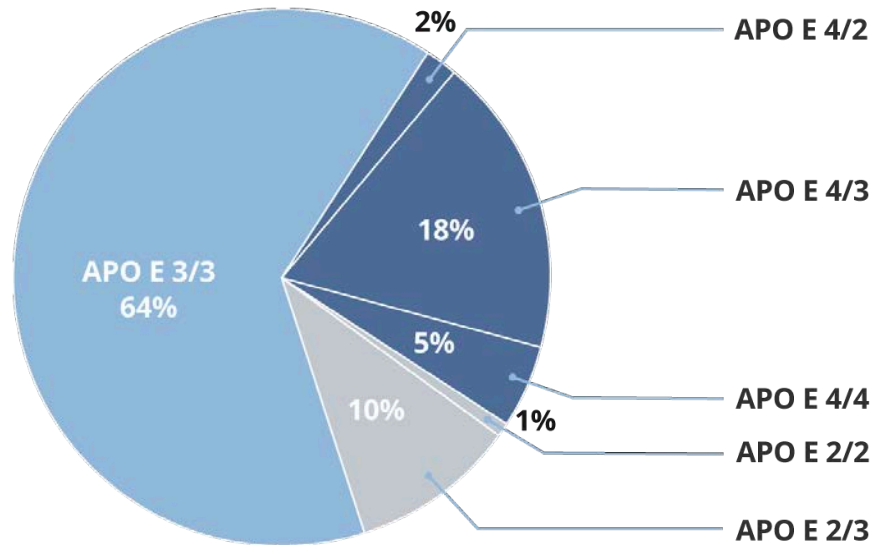
Jun. 28, 2002 (hg12)

Práctica: raw format de microarrays

- Descargar datos raw del tipo "23andMe" desde Harvard PGP
- Descomprimir (si es necesario) y visualizar su contenido
- Consultar en dbSNP alguno de los rsIDs
- Convertir a formato VCF
- Visualizar con IGV

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Alzheimer y APOE



Genotype	E2/E2	E2/E3	E2/E4	E3/E3	E3/E4	E4/E4
Disease Risk	40% less likely	40% less likely	2.6 times more likely	Average risk	3.2 times more likely	14.9 times more likely

APOE Allele	rs429358	rs7412
$\epsilon 2/\epsilon 2$	TT	TT
$\epsilon 2/\epsilon 3$	TT	TC
$\epsilon 2/\epsilon 4$	TC	TC
$\epsilon 3/\epsilon 3$	TT	CC
$\epsilon 3/\epsilon 4$	TC	CC
$\epsilon 4/\epsilon 4$	CC	CC

Práctica: APOE genotipo

- Genotipado de
 - rs429358
 - rs7412
- Visualización con IGV del gen APOE y los dos rsIDs anteriores

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BRCA1 y BRCA2

- Los genes BRCA1 y BRCA2 codifican proteínas que son fundamentales para que las células reparen el ADN dañado
- Las mutaciones específicas heredadas en estos genes aumentan el riesgo de varios tipos de cáncer, en particular el de mama y el de ovario
- El riesgo se ve influido por la localización de las mutaciones dentro de los genes BRCA y el alcance de los antecedentes familiares de cáncer de mama

Estadísticas BRCA1, BRCA2

BRCA1/2 associated cancer risks.

Cancer Type	General Population Risk	Mutation Risk	
		BRCA1	BRCA2
Breast	12%	50%-80%	40%-70%
Second primary breast	3.5% within 5 years Up to 11%	27% within 5 yrs	12% within 5 yrs 40%-50% at 20 yrs
Ovarian	1%-2%	24%-40%	11%-18%
Male breast	0.1%	1%-2%	5%-10%
Prostate	15% (N. European origin)	<30%	<39%
	18% (African Americans)		
Pancreatic	0.50%	1%-3%	2%-7%



EL PAÍS

La revelación de Angelina Jolie mueve el parqué de Wall Street

Las acciones de Myriad Genetics, que desarrolla la prueba que detecta si se puede padecer cáncer de mama, se dispararon un 15% tras la confesión de la actriz



SANDRO POZZI

Nueva York - 10 JUN 2013 - 18:26 CEST

PATIENTS PROVIDERS PRODUCTS ABOUT CONTACT

SEARCH

GENETIC INSIGHTS

February is National Cancer Prevention Month

If you have a family history of cancer, you can understand your risk. Get started today by taking the Hereditary Cancer Quiz.

[TAKE THE QUIZ](#)

Genetic insights for your health

BRCAnalysis

BRACAnalysis®

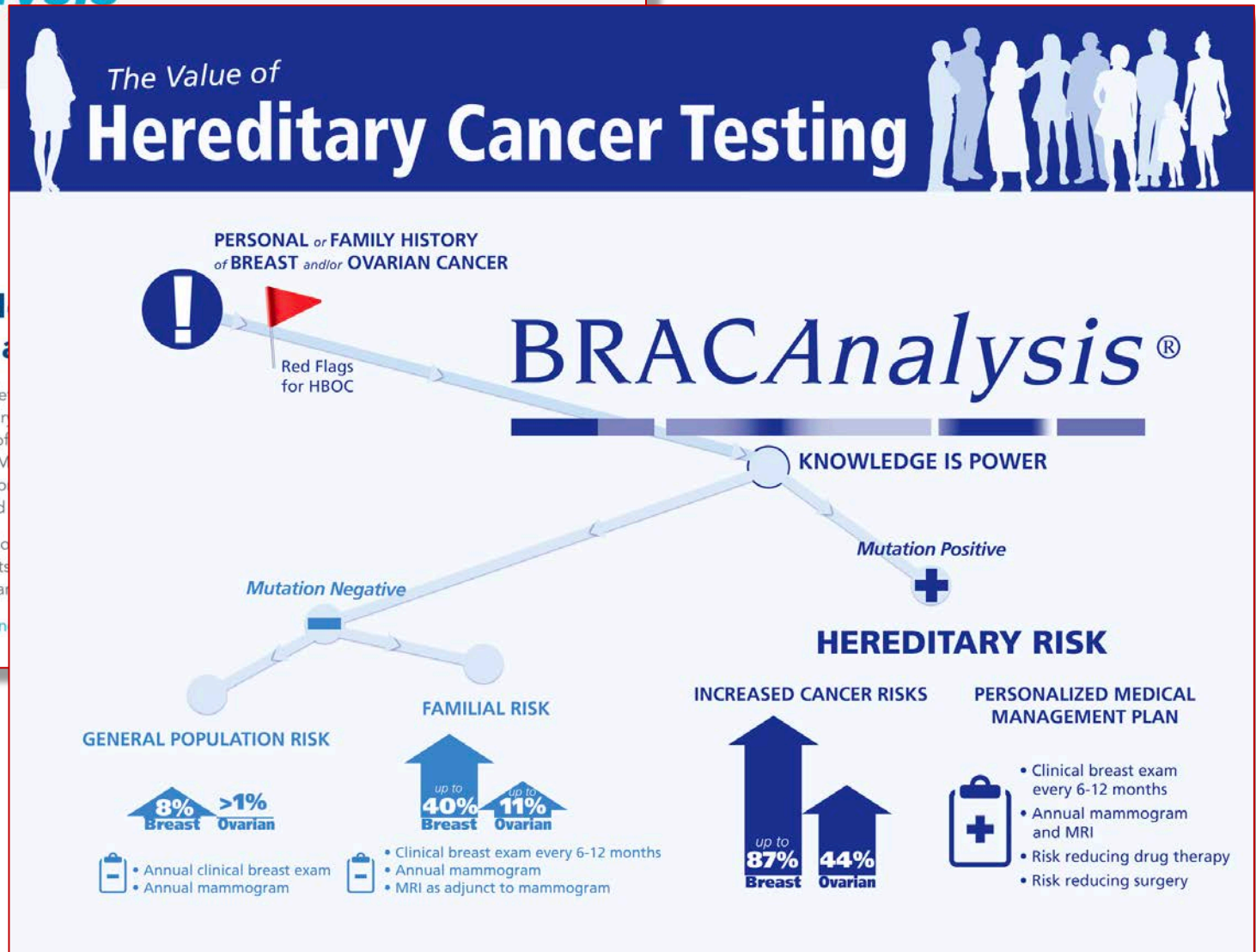
BRACAnalysis®

BRACAnalysis®: Hereditary Breast and Ovarian Cancer

BRACAnalysis® is a genetic test that determines if you are responsible for the majority of hereditary breast and ovarian cancer. The *BRCA1* or *BRCA2* gene have risks of developing ovarian cancer by age 70. Mutations in these genes increase the risk of developing a second primary cancer, a 65 percent chance of developing a second primary cancer.

BRACAnalysis requires only a simple blood sample to determine if you have a *BRCA1* or *BRCA2* mutation. Knowing the results of the test can help you understand the onset of cancer or detect it at an early stage.

Myriad has developed a Hereditary Cancer genetic testing using BRACAnalysis.



BRCA1, BRCA2 ubicación

Genomic Locations for BRCA1 Gene

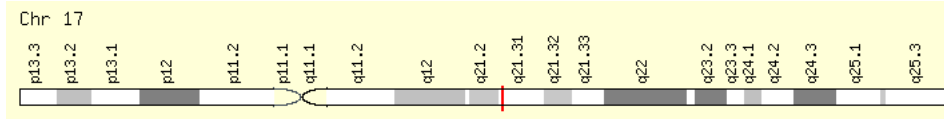
Genomic Locations for BRCA1 Gene

chr17:43,044,295-43,170,245 (GRCh38/hg38)

Size: 125,951 bases Orientation: Minus strand

chr17:41,196,312-41,277,500 (GRCh37/hg19)

Size: 81,189 bases Orientation: Minus strand



Genomic Locations for BRCA2 Gene

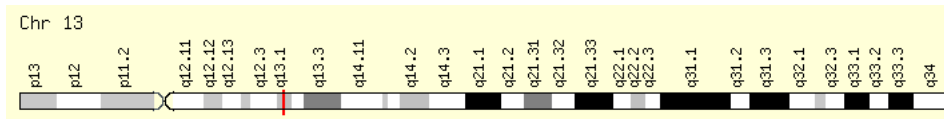
Genomic Locations for BRCA2 Gene

chr13:32,315,086-32,400,268 (GRCh38/hg38)

Size: 85,183 bases Orientation: Plus strand

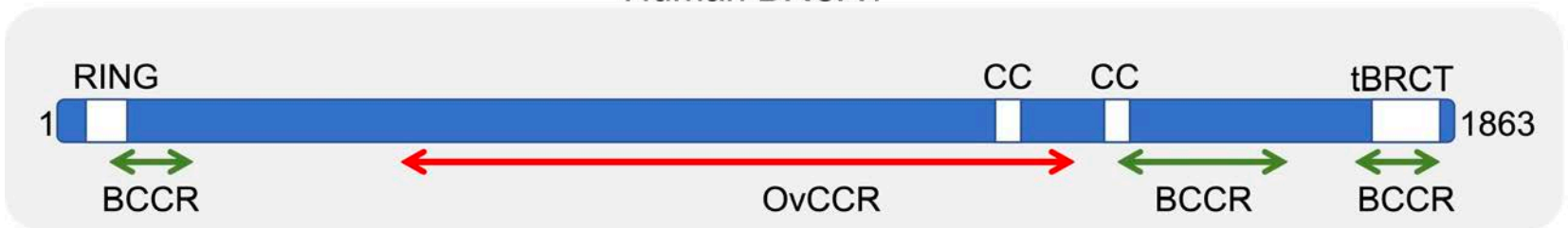
chr13:32,889,611-32,973,809 (GRCh37/hg19)

Size: 84,199 bases Orientation: Plus strand

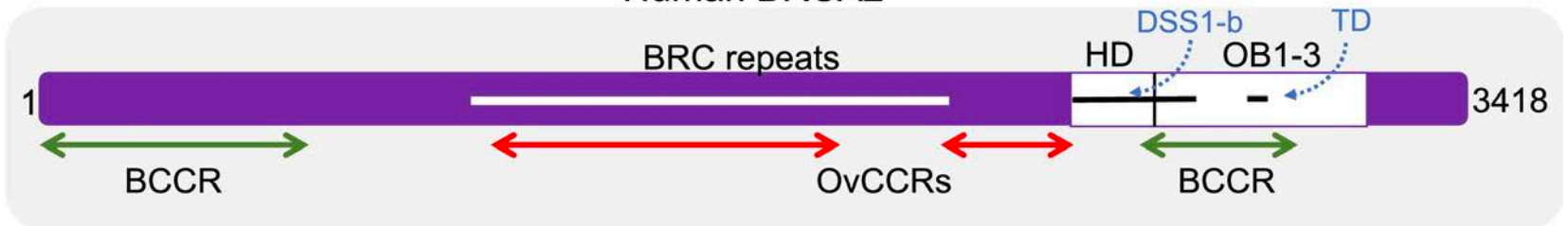


BRCA1, BRCA2

Human BRCA1



Human BRCA2



BRCA Exchange

 **BRCA Exchange** [Summary View](#)

[HOME](#) [VARIANTS](#) [COMMUNITY](#) [HELP](#) [MORE ▾](#)

search for "c.1105G>A", "brca1" or "IVS7+1037T>C"



The BRCA Exchange aims to advance our understanding of the genetic basis of breast, ovarian, pancreatic and other cancers by pooling data on BRCA1/2 genetic variants and corresponding clinical data from around the world. Search for BRCA1 or BRCA2 variants above.

This website is a product of the BRCA Challenge project, a driver project of the Global Alliance for Genomics and Health.



What is the BRCA Exchange?

Video produced by [Kindea Labs](#)



How Does the BRCA Exchange Benefit Patients?

Video produced by the [Global Alliance for Genomics and Health](#) and the [Broad Institute](#)

Práctica: genotipado de BRCA1 y BRCA2

- Identificar las variantes reportadas que afectan a BRCA1 y BRCA2
- Pasos:
 - Detectar las variantes presentes en las coordenadas genómicas correspondientes a cada gen
 - Verificar si las variantes son patogénicas
 - ANNOVAR
 - BRCA Exchange

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- Interpretación: Odds ratio

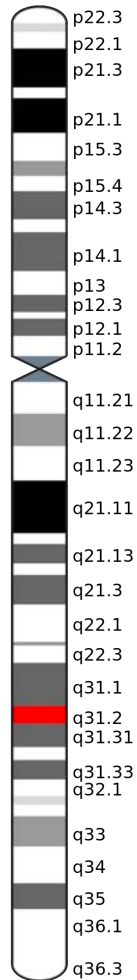
Fibrosis quística

- Enfermedad genética
- Herencia autosómica recesiva
- Afecta a pulmones y en menor medida páncreas, hígado e intestino

PAIS	INCIDENCIA
Dinamarca	1/4700
España	1/7213
Finlandia	1/25000
Francia	1/4000
Francia (Bretaña)	1/2913
Holanda	1/3600
Irlanda	1/1461
Italia	1/2730
Noruega	1/6500
Reino Unido	1/2400
Suecia	1/4000
Suiza	1/2000

Gen CFTR

Chromosome 7



- Símbolo gen: CFTR
- Nombre proteína: Cystic Fibrosis Transmembrane conductance Regulator
- Cromosoma 7, región 7q31
- 189 kb
- 27 exones
- RefSeq: NP_000483
- Uniprot: P13569
- Proteína: 1480 aa

Base de datos de mutaciones: estadísticas



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CFMDB Statistics

There are currently **2103** mutations listed in this CFTR mutation database.

Statistics by mutation type:

Mutation Type	Count	Frequency %
Missense	815	38.75
Frameshift	342	16.26
Splicing	230	10.94
Nonsense	177	8.42
In frame in/del	43	2.04
Large in/del	59	2.81
Promoter	17	0.81
Sequence variation	269	12.79
Unknown	151	7.18

Frecuencia de mutaciones (I)

Journal of
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ORIGINAL ARTICLE | VOLUME 13, ISSUE 4, P403-409, JULY 01, 2014

The relative frequency of *CFTR* mutation classes in European patients with cystic fibrosis

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Abstract

Abstract

Keywords

Introduction

Methods

Results

Discussion

Acknowledgments

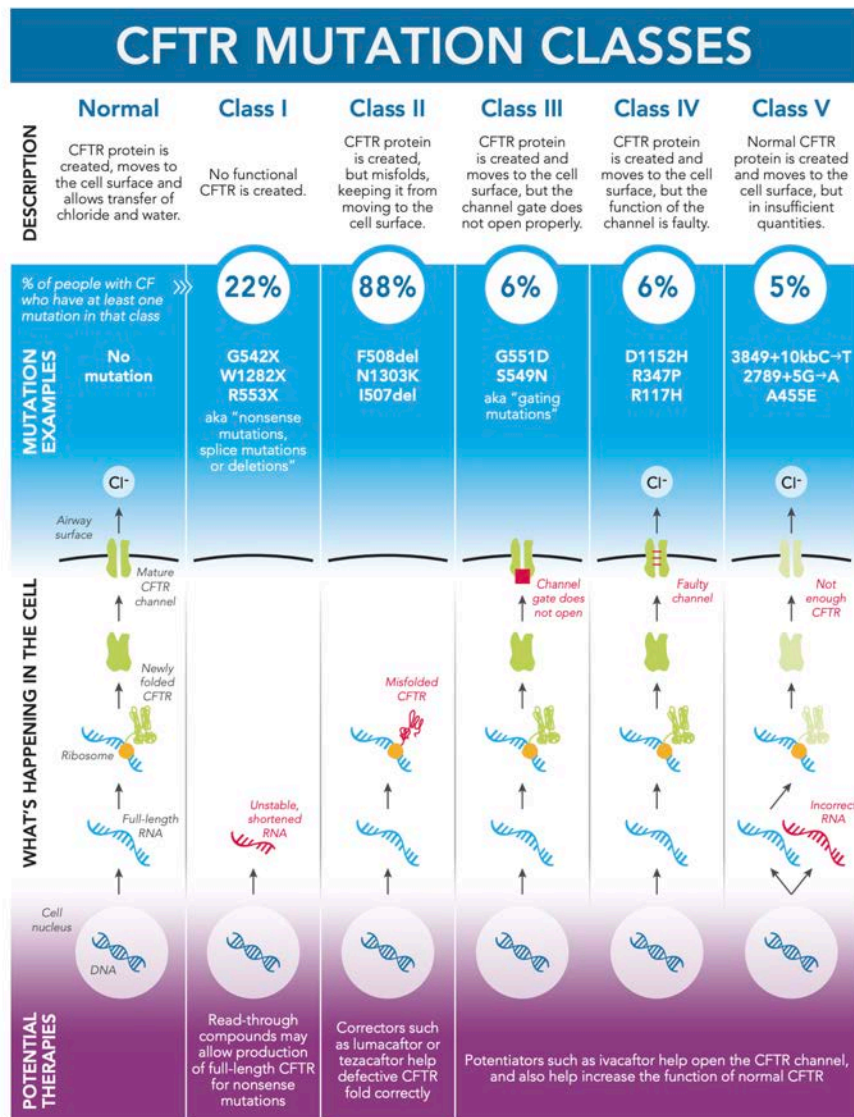
Supplementary data

References

More than 1900 different mutations in the *CFTR* gene have been reported. These are grouped into classes according to their effect on the synthesis and/or function of the CFTR protein. CFTR repair therapies that are mutation or mutation class specific are under development. To progress efficiently in the clinical phase of drug development, knowledge of the relative frequency of *CFTR* mutation classes in different populations is useful. Therefore, we describe the mutation class spectrum in 25,394 subjects with CF from 23 European countries.

In 18/23 countries, 80% or more of the patients had at least one class II mutation, explained by F508del being by far the most frequent mutation. Overall 16.4% of European patients had at least one class I mutation but this varied from 3 countries with more than 30% to 4 countries with less than 10% of subjects. Overall only respectively 3.9, 3.3 and 3.0% of European subjects had at least one mutation of classes III, IV and V with again great variability: 14% of Irish patients had at least one class III mutation, 7% of Portuguese patients had at least one class IV mutation, and in 6 countries more than 5% of patients had at least one class V mutation.

Frecuencia de mutaciones (II)



KNOW YOUR MUTATIONS: A CFTR Mutation Fact Sheet



Cystic fibrosis is caused by mutations, or changes, in the CFTR gene. This gene provides the code that tells the body how to make the cystic fibrosis transmembrane conductance regulator (CFTR) protein. The protein controls the salt and water balance in the lungs and other tissues. All people have two copies of the CFTR gene, and there must be mutations in both copies to cause CF. More than 1,700 mutations of the CFTR gene have been identified. Although some are common, others are rare and found in only a few people.

CFTR mutations are grouped into classes based on the way the mutations affect the CFTR protein. The reverse side of this sheet shows the most common CFTR mutation classes. In the future, mutations may also be classified by "theratype," meaning which type of CFTR modulator therapy they respond to best. This is because mutations within the same class may

respond to therapies differently, and not every mutation can be neatly assigned to one mutation class.

Certain types of CFTR mutations are associated with different disease complications. For example, some mutations are more likely to affect the pancreas than others. However, this correlation is not perfect, and knowing an individual's CFTR mutations cannot always tell you how severe that person's CF symptoms will be.

Although the potential therapies described on this sheet can be very effective for some people with CF, others may not experience the exact same benefit. Researchers continue to work in the lab and in clinical trials to find the best therapeutic approaches to target specific CFTR mutations or classes of mutations to improve the health of all individuals living with CF.

How is CFTR made?

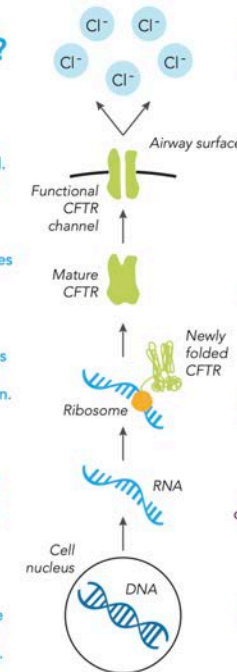
Once at the cell surface, the CFTR protein functions as a chloride channel. This channel helps maintain the right balance of fluid in the airways.

Once complete, the CFTR protein moves through the cell to the cell surface. This process is called trafficking.

Ribosomes are tiny molecular machines that read the instructions in the RNA and use them to make the CFTR protein. This process is called translation.

RNA is created by matching the coded instructions in the DNA. This process is called transcription. RNA acts as a template to make proteins.

DNA in the cell nucleus provides instructions to make proteins. The CF gene contains instructions to make the cystic fibrosis transmembrane conductance regulator (CFTR) protein.



Potential therapies for CFTR mutations

Potentiators are drugs that help open the CFTR channel at the cell surface and increase chloride transport.

Correctors are drugs that help the defective CFTR protein fold properly so that it can move to the cell surface.

Read-through compounds aim to allow full-length CFTR protein to be made, even when the RNA contains a mutation telling the ribosome to stop.

RNA therapies aim to either fix the incorrect instructions in defective RNA, or provide normal RNA directly to the cell.

Gene-editing techniques aim to repair the underlying genetic defect in the CF gene DNA. Gene replacement techniques aim to provide a correct copy of the CF gene.

Práctica: genotipado de CFTR

- Identificar las variantes reportadas que afectan CFTR
- Pasos:
 - Detectar las variantes presentes en las coordenadas genómicas correspondientes al gen
 - Verificar si las variantes son patogénicas
 - ANNOVAR

- Patrones de herencia
- Microarray de genotipado
- Bases de datos de variantes
- Direct-to-Consumer Genomic Testing
- Personal Genome Project
- Enfermedades monogénicas
 - Alzheimer y APOE
 - Cáncer de mama y útero y BRCA1, BRCA2
 - Fibrosis quística y CFTR
- Interpretación: Odds ratio

Sentido común e interpretación cautelosa

- Preguntas clave
 - ¿Es realmente probable que el rasgo/enfermedad sea producido por un solo SNP?
 - Probablemente haya que hacer algún estudio adicional
 - ¿Es correcto y reproducible?
 - Número de estudios/publicaciones relacionadas con el SNP
 - ¿En qué medida me va a afectar como individuo?
 - Tamaño de efecto
 - Necesitamos la estadística

Odds ratio (OR)

- Si encontramos que un SNP tiene un $OR=1.3$ para un alelo G de un accidente cerebrovascular significa:
 - Para cada alelo G es 1.3 veces más probable que una persona esté en el grupo de personas que tiene accidentes cerebrovasculares
- Valores típicos:
 - 1.1-1.5
- Valores no tan típicos:
 - $OR=3.5$ APOE en Alzheimer
 - Tabaquismo $OR=16$, para provocar cáncer de pulmón
 - No todo es la composición genética del individuo: ambiente y hábitos

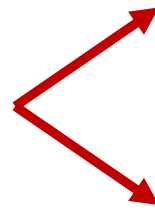
DESVÍO: Traducciones de *odds ratio*

- desigualdad relativa
- razón de productos cruzados
- producto cruzado
- cociente de probabilidades relativas
- oportunidad relativa
- razón de oposiciones
- razón de momios
- razón del producto cruzado
- razón de ventajas
- relación de probabilidad
- razón de probabilidad
- relación impar
- razón de posibilidades

Tabla de contingencia 2x2

	Alelo riesgo si	Alelo riesgo no	Total
Caso (enfermo)	<i>a</i>	<i>b</i>	<i>a+b</i>
Control (sano)	<i>c</i>	<i>d</i>	<i>c+d</i>
Total	<i>a+c</i>	<i>b+d</i>	<i>a+b+c+d</i>

Simétrico


$$OR = \frac{a/c}{b/d} = \frac{ad}{bc}$$
$$OR = \frac{a/b}{c/d} = \frac{ad}{bc}$$

$$SE(\ln(OR)) = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$$

OR: Ejemplo 1 (parte 1)

Enfermedad cardiovascular
rs4977574

	Alelo riesgo si	Alelo riesgo no	Total
Enfermo	8000	7000	15000
Sano	7000	8000	15000
Total	15000	15000	30000

$$OR = \frac{8000/7000}{7000/8000} = \mathbf{1.31}$$

$$SE(\ln(OR)) = \sqrt{\frac{1}{8000} + \frac{1}{7000} + \frac{1}{7000} + \frac{1}{8000}}$$

$$CI(95)_{lower} = e^{\ln(OR) - 1.96 * SE} = \mathbf{1.25}$$

$$CI(95)_{upper} = e^{\ln(OR) + 1.96 * SE} = \mathbf{1.37}$$

OR: Ejemplo 1 (parte 2)

Se sabe que aproximadamente el 0.1% de la población entre 35 y 74 años muere cada año a causa de una enfermedad coronaria

Ahora hacemos el muestro aleatoriamente en individuos entre 35 y 74 años

¿Qué encontraríamos para tener un OR = 1.3?

	Alelo riesgo si	Alelo riesgo no	Total
Enfermo	6	5	10
Sano	4700	5300	10000
Total	4706	5305	10011

OR = 1.35 (CI95: 0.41-4.44)

$$11/10011 = 0.11\%$$

$$6/4706 = 0.13\%$$

Conclusión: si tengo el alelo de riesgo tengo un 0.13% de probabilidad de estar enfermo, en lugar de 0.11%.

Es mejor decir que el OR es 1.3, que decir que se tiene un 30% de incremento de riesgo

OR: Ejemplo 2. Efecto alto.

Fumador y cáncer de pulmón

	Fumador	No fumador	Total
Enfermo	647	2	649
Sano	622	27	649

OR = **14.04** (CI95: 3.3-59.3)

Preocupante:

- OR > 5
- “complete disease association”
- “completely penetrant”

OR: Ejemplo 3. Gene BRCA2

	BRCA2 alelo	NO BRCA2 alelo
Cáncer de mama	3k	110k
No cáncer de mama	2k	890k

OR = **12.1** (CI95: 2.0-73.4)

3/2 → 50% más probable cáncer de mama si alelo
(3+2)/(3+110+2+890) → 0.5% frecuencia de mutación BRCA2
(110+3)/(3+110+2+890) → 11% prevalencia cáncer de mama

Cálculos de OR con Python

```
import numpy as np
from sys import argv
from scipy.stats import norm

a = int(argv[1])
b = int(argv[2])
c = int(argv[3])
d = int(argv[4])

oddsratio = (a*d)/(b*c)
logodd = np.log(oddsratio)
selogodd = np.sqrt(1/a + 1/b + 1/c + 1/d)
#ci_low_95 = np.exp(logodd-1.96*selogodd)
#ci_upp_95 = np.exp(logodd+1.96*selogodd)
ci_low_90 = np.exp(logodd-norm.ppf(.95)*selogodd)
ci_upp_90 = np.exp(logodd+norm.ppf(.95)*selogodd)
ci_low_95 = np.exp(logodd-norm.ppf(.975)*selogodd)
ci_upp_95 = np.exp(logodd+norm.ppf(.975)*selogodd)
ci_low_99 = np.exp(logodd-norm.ppf(.995)*selogodd)
ci_upp_99 = np.exp(logodd+norm.ppf(.995)*selogodd)
print ('Odds ratio = %.2f' %oddsratio)
print ('CI (90%%) = %.2f - %.2f' % (ci_low_90,ci_upp_90))
print ('CI (95%%) = %.2f - %.2f' % (ci_low_95,ci_upp_95))
print ('CI (99%%) = %.2f - %.2f' % (ci_low_99,ci_upp_99))
```


Genotipado con microarrays

Máster en Sistemas Inteligentes

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Departamento de Informática

